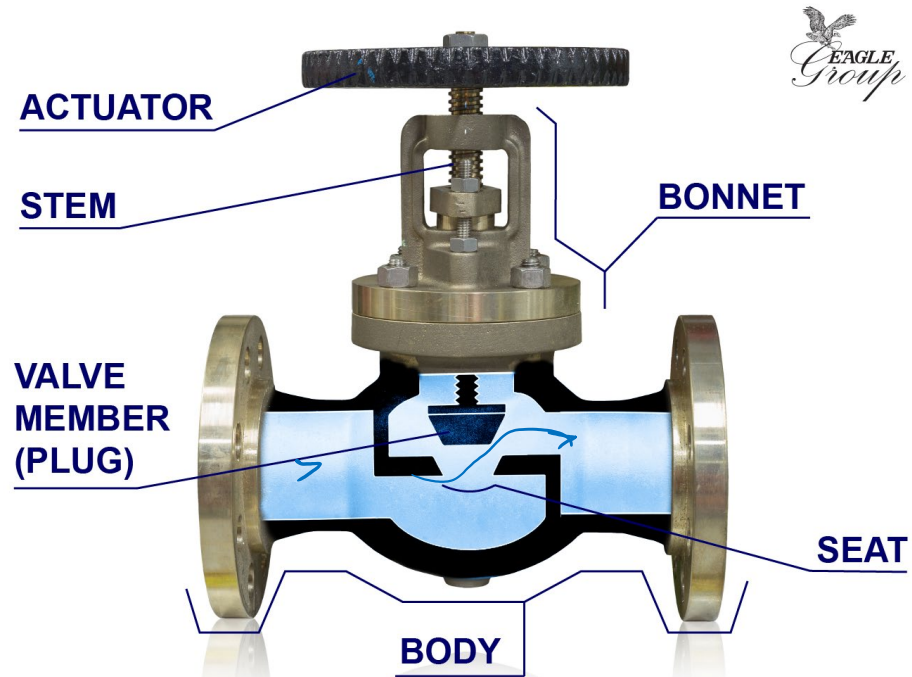




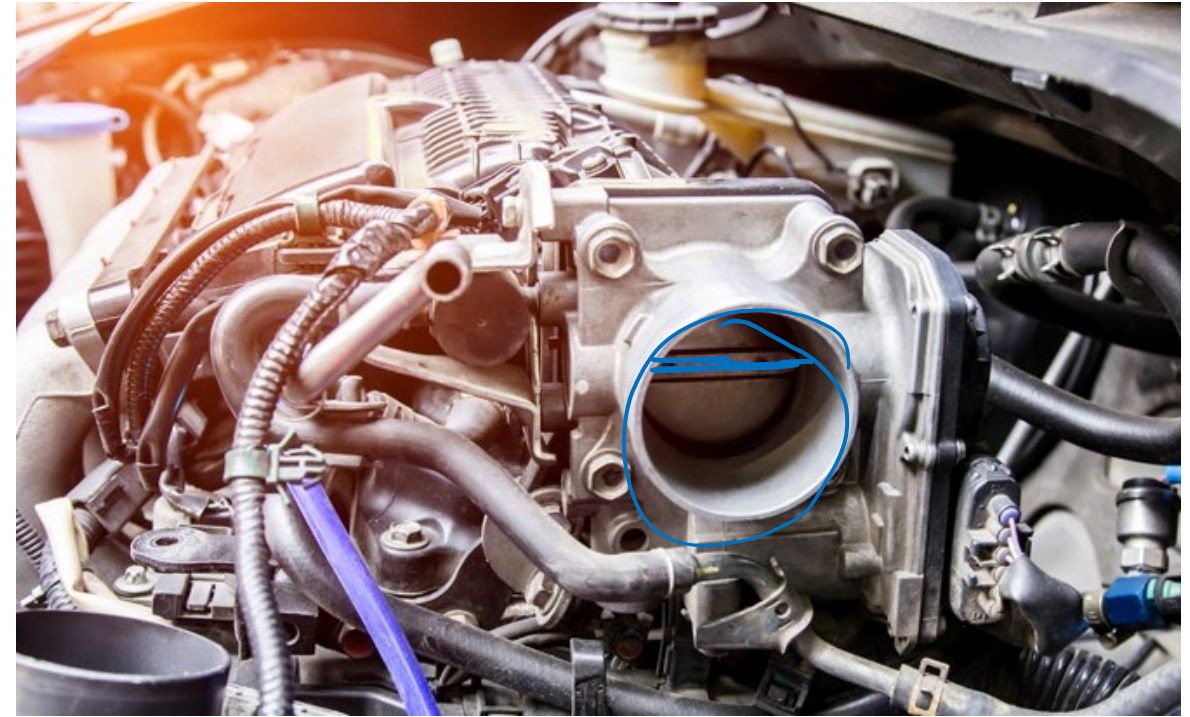
ME 300: Engineering Thermodynamics

Throttles

What is a throttle?



<https://blog.eaglegroupmanufacturers.com/anatomy-of-industrial-valves>



<https://www.autoguide.com/best-throttle-body-cleaner>



sintered plate
filter

<https://www.sinteredfilter.org/wiremesh/sintered-stainless-steel-powder-filter-element.html>

Throttle assumptions

$$\rightarrow \dot{W} = 0 \quad \text{passive}$$

$$\dot{Q} = 0 \quad \text{adiabatic}$$

$$\Delta k = 0$$

$$\Delta p e = 0$$

$$\dot{m} (\cancel{\Delta h} + \cancel{\alpha \cancel{v_e^2}} + \cancel{\alpha \cancel{p_e}}) = \cancel{0} - \cancel{\dot{W}}$$

$$\Delta h = 0$$

$$\boxed{h_2 = h_1}$$

$$h = \underbrace{u + Pv} \quad P_2 < P_1$$